



EEG effective connectivity during the first year of life mirrors brain synaptogenesis, myelination, and early right hemisphere predominance

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ABSTRACT

Introduction: The maturation of electroencephalogram (EEG) effective connectivity in healthy infants during the first year of life is described.

Methods: *Participants:* A cross-sectional sample of 125 healthy at-term infants, from 0 to 12 months of age, underwent EEG in a state of quiet sleep. *Procedures:* The EEG primary currents at the source were described with the sLoreta method. An unmixing algorithm was applied to reduce the leakage, and the isolated effective coherence, a direct and directed measurement of information flow, was calculated.

Results and Discussion: Initially, the highest indices of connectivity are at the subcortical nuclei, continuing to the parietal lobe, predominantly the right hemisphere, then expanding to temporal, occipital, and finally the frontal areas, which is consistent with the myelination process. Age-related connectivity changes were mostly long-range and bilateral. Connections increased with age, mainly in the right hemisphere, while they mainly decreased in the left hemisphere. Increased connectivity from 20 to 30 Hz, mostly at the right hemisphere. These findings were consistent with right hemisphere predominance during the first three years of life. Theta and alpha connections showed the greatest changes with age. Strong connectivity was found between the parietal, temporal, and occipital regions to the frontal lobes, responsible for executive functions and consistent with behavioral development during the first year. The thalamus exchanges information bidirectionally with all cortical regions and frequency bands.

Conclusions: The maturation of EEG connectivity during the first year in healthy infants is very consistent with synaptogenesis, reductions in synaptogenesis, myelination, and functional and behavioral development.

1. Introduction

During the first year of life, an infant's brain experiences many structural and physiological changes. Several studies have reported age-related developmental effects on magneto electroencephalogram (MEG) (Okada et al., 2016) and electroencephalogram (EEG) signals during the first year of life (Otero et al., 2011; Bosch-Bayard et al., 2012; Liu et al., 2015; Schaworonkow and Voytek, 2021). However, although EEG connectivity has been studied in children, adolescents, and adults (Boersma et al., 2011; Kurth et al., 2010), there is scarce literature about its characteristics during normal infants' first year of life. References

concerning connectivity in infants have been referred to preterm infants (Jennekens et al., 2012; Meijer et al., 2016, 2014; Niemark et al., 2010), neonates (Duffy et al., 2003; González et al., 2011), low birth weight infants (Grieve et al., 2008), growth restricted infants (Cohen et al., 2018), or infants at risk for autism spectrum disorder (Righi et al., 2014). Other studies have used magnetic resonance imaging in normal subjects (De Asis-Cruz et al., 2020; Kulikova et al., 2015; Liu et al., 2015) and in patients with neonatal brain injury (Smyser et al., 2019).

The study of EEG oscillations according to their frequency has increased since different references, which will be mentioned later, have proposed a specific role for some of them concerning communication.

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The alpha rhythm has been shown to be related to inhibition (Jensen and Mazaheri, 2010; Klimesch, 2012). The alpha rhythm, coupled with gamma activity, produces cyclic inhibition to increase the signal-to-noise ratio (Modolo et al., 2020). Therefore, the alpha rhythm plays an essential role in controlling “top-down” information, influencing information processing and the interplay of information (Sadaghiani and Kleinschmidt, 2016). This role is comparable across different sensory modalities (visual, auditory, and somatosensory) and the motor system. It has also been shown that temporal coding organized by coupled alpha and gamma oscillations prioritizes visual processing (Jensen and Mazaheri, 2010). Cross-frequency coupling of theta and gamma oscillations is a possible substrate of an information channel through distant regions during consciousness that opens and closes (theta rhythm) and that processes and transfers (gamma) information in the brain (Modolo et al., 2020).

Coherent brain rhythms represent a core mechanism for sculpting temporal coordination of neural activity in the brain-wide network (Wang, 2010). Brain rhythmic activity in one area is likely to be communicated to another and thus generate a temporal linkage in the rhythmicity of the two structures. Coherent oscillations are thought to play a role in well-timed coordination and communication between neural populations simultaneously engaged in cognitive processes (Wang, 2010).

The synchronization of oscillatory activity is relevant for the development of cortical circuits, as demonstrated by the involvement of neural synchrony in synaptic plasticity and changes in the frequency and synchronization of neural oscillations during development (Uhlhaas et al., 2009). The emergence of discrete bands in infancy and childhood may reflect the maturation of distinct cortical generators producing the observed rhythms (Bollimunta et al., 2011) and the processes of synaptogenesis and pruning (Kurth et al., 2010; Uhlhaas et al., 2009).

Early studies on the structural neurodevelopment of the cortex in monkeys (Rakic et al., 1986) and in humans (Huttenlocher, 1979; Huttenlocher and Dabholkar, 1997) demonstrated that the primate brain overproduces synapses early in development. These studies reported that the peak of synaptic overproduction in the human visual cortex occurred in approximately the fourth postnatal month. As a result, a full-term newborn brain has many more synapses than an adult brain. However, this period of synaptogenesis is followed by a period of synapse reduction. It has also been shown that the time course of these periods varies immensely according to the brain regions in humans (Thompson and Nelson, 2001; Lebenberg et al., 2019). The auditory cortical areas showed the peak of synaptic overproduction at approximately 2–3 postnatal months and the areas of receptive language (angular gyrus) at 7–8 postnatal months until 3–4 years of age. The areas of speech production began synaptogenesis around the eighth month and continued until the age of 4 years. The frontal lobe is the last lobe to initiate synaptogenesis at 8–12 months, and the peak of synaptogenesis is at 11–12 months. According to Paterson et al. (2006), the period between 8 and 12 months is crucial in the development of the frontal lobes. Additionally, Thompson and Nelson (2001) considered that the early postnatal months constitute a sensitive period for the development of executive functions that are characteristic of the frontal lobes. Another important change during postnatal neural development is the increase in size and complexity of the dendritic trees of most neurons. While the extent and reach of a cell’s dendritic arbor may increase dramatically, it also often becomes more specific and specialized during postnatal neural development (Baars and Gage, 2013).

Myelination begins at birth in the base of the brain, with the pons and cerebellar peduncles progressing to optic radiation, the splenium, and the corpus callosum (Paus et al., 2001). Although almost all the cortices begin myelination at 2–3 months, this process ends at different times in different regions: myelination in the auditory and visual cortices ends at 18–24 months, in Broca’s area at 5 years and in the prefrontal cortex at 9 years of age. Additionally, maturational asynchrony exists across the bundles: the spinothalamic tract, optic radiations, corticospinal tract, and fornix have the most advanced maturation, while the superior lon-

gitudinal and arcuate fasciculi, anterior limb of the internal capsule and external capsule have the most delayed maturation (Dubois et al., 2014). Chi et al. (1977), investigating 507 brains from infants of 10–44 weeks gestational age, found left-right asymmetries of the transverse temporal gyri, Sylvian fissures, and planum temporale. In general, the right cerebral hemisphere showed gyral complexity earlier than the left cerebral hemisphere. Changes in cerebral blood flow measured at rest using dynamic single-photon emission computed tomography in children between 1 and 3 years of age showed a right hemispheric predominance; these asymmetries shifted to the left after 3 years (Chiron et al., 1997).

In this work, EEG recordings were used to assess the development of functional connectivity between different brain areas in healthy infants in their first year of life. Our hypotheses are as follows: (1) the connectivity between different regions will show age-related changes during the first year of age. (2) Connectivity between regions in all EEG frequency bands will increase with age according to myelination maturation and synaptogenesis. (3) Connectivity between regions in all EEG frequency bands will decrease with age according to the period of synapse reduction. (4) A right hemispheric predominance will be observed in the EEG connectivity between different regions.

2. Materials and methods

2.1. Ethics

The Ethics Committee of the Instituto de Neurobiología of the Universidad Nacional Autónoma de México approved this study (protocol number 023. H), which also complied with the Ethical Principles for Medical Research Involving Human Subjects established by the Declaration of Helsinki. All infants and children were volunteers, and informed consent was obtained from the parents of all study participants.

2.2. Participants

The sample was obtained from two public hospitals in Querétaro: the “Hospital de Especialidades del Niño y la Mujer” and the “Hospital del Instituto Mexicano del Seguro Social”, which are the hospitals that receive the largest number of children in the state of Querétaro. This made the sample representative of the Mexican population. Mothers of the children that had no history of complications during pregnancy and childbirth, were asked to go to the Neurodevelopmental Research Unit of the Institute of Neurobiology (UNAM-Mexico) and they were told that they were going to have several free evaluations such as the study by a pediatrician expert in Neurology, the registration of the electroencephalogram, of the visual and auditory evoked potentials and of a scale that measured neurodevelopment.

From all evaluated infants, 125 of them who fulfilled the inclusion criteria, were selected: 74 males and 51 females (74/51/125), from birth to 1 year of age. The mean age was 5.4 months (± 3.12). By months, the sample is distributed as follows: 0–3 months, 18/15/33; 3–6 months, 29/14/33; 6–9 months, 13/14/27; 9–12 months, 14/8/22. This distribution shows that there was an adequate representation of all ages along the first year of life, being the ages from 0 to 6 months slightly higher represented.

The inclusion criteria were normal delivery at term, weight at birth between 2500 and 3900 g (this weight range characterizes well our healthy newborns, since Mexican persons are in average smaller than Anglo-Saxon individuals. On the other hand, in Mexico there is a high incidence of diabetes and therefore there is a great control of glycemia in pregnant women and the weight of newborns), Apgar ≥ 8 at the first minute and ≥ 9 at the fifth minute after birth, no antecedents of pre- or perinatal risk factors for brain damage, normal pediatric examination results, and scores between normal limits on the neurodevelopmental scale (Bayley II, 2003). Bayley’s scores varied according to the age of evaluation, although within normal limits. Mean scores for the 125 subjects were for MDI = 102.28 ± 11.88 and for PDI = 108.47 ± 12.56 .

2.3. Recordings

EEG recordings were obtained from infants while they were in phase II sleep. Infants were in his or her mother's lap in a dimly lit room with acoustic isolation. An EEG was recorded from 19 leads (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, and Pz) using the 10/20 International System with linked ear lobes as the reference. The amplifier bandwidth was set between 0.5 and 100 Hz. The EEG was sampled every 5 ms using a MEDICID IV System and edited offline. The Bayley evaluation was not exactly at the same time than EEG acquisition. Differences in the interval between them were not greater than 5 weeks.

2.4. EEG analysis

The EEG epochs for the analysis were selected by visual inspection of four expert neurophysiologists (TH, GAO, JRG and TF). In each recording, they carefully selected and verified 24 artifact-free and quasi-stationary EEG segments of 2.56 s during physiologically quiet sleep. This procedure is standard in the clinical environment and very reliable when dealing with 19 channels recordings, if performed by well-trained neurophysiologists, as it is our case.

Special attention was devoted to avoiding transient events in the selected windows. It is well known that the EEG is not stationary, but it has also been shown that is easy to select epochs of quasi-stationary epochs of around 2–4 s length. In our case, the 2.56 s epochs comprise 512 time points (5 ms sampling period), appropriate for the estimation of the multivariate autoregressive model used by iCoh.

Not any automatic procedures for cleaning the signals were used, since they are neither adequate nor reliable for the purpose of connectivity analysis. Most of such procedures are based either in interpolations or components decomposition and signal reconstruction after eliminating artifactual components. Such methods introduce linear correlations among the signals, generating artificial patterns of connectivity.

For our purpose of analyzing functional connectivity, the EEG signals at the scalp cannot be used directly since they are affected by the volume conduction effect (Nunez and Srinivasan, 2009) and produce spurious connections. Instead, the scalp signals were projected to the source level by applying an inverse solution method. However, inverse methods have low resolution, and the signals at the sources are still affected by the mixing (leakage) problem, which depends on the resolution matrix of the adopted inverse method (Biscay et al., 2018).

2.4.1. Unmixed estimation of the primary current at the sources

Biscay et al. (2018) proposed an unmixing algorithm for linear inverse methods based on the analysis of the resolution matrix of the adopted inverse method. They also demonstrated that the unmixing algorithm achieves the same results for all the linear inverse methods, making their results independent of the inverse method used. This property makes the unmixing algorithm particularly attractive since the results are more robust and reproducible.

Another characteristic of the method described in Biscay et al. (2018) is that it works perfectly well to unmix not only single-source signals but also multisource regions of interest (ROIs) that may or may not cover the whole cortex. In that case, the unmixed signal of the ROIs is the unmixed average signal of the sources. The only requirement for the method to work properly is that the ROIs must be disjointed.

However, one limitation is that the unmixing of the EEG source signals is limited by the number of EEG electrodes recorded at the scalp because only the number of electrodes (-1) signals are linearly independent and could therefore be unmixed. The mentioned authors already showed that starting the unmixing procedure from the signals estimated by the inverse method produce better results than directly estimating only the small number of signals which will be involved in the connectivity analysis, since the inverse method itself produce an amount of

Table 1

Brain regions with the highest global connectivity index (GCI) in order of numeration.

1. Left amygdala	18. Right Heschl's gyrus
2. Right hippocampus	19. Left middle cingulum
3. Left thalamus	20. Right anterior cingulum
4. Right thalamus	21. Left middle temporal gyrus
5. Right caudate	22. Left supplementary motor area
6. Left Rolandic operculum	23. Right inferior occipital gyrus
7. Left Heschl's gyrus	24. Left gyrus rectus
8. Right putamen	25. Right middle temporal gyrus
9. Left parahippocampal gyrus	26. Right superior temporal gyrus
10. Right olfactory bulb	27. Right inferior frontal operculum
11. Right parahippocampal gyrus	28. Right supramarginal gyrus
12. Left olfactory bulb	29. Right middle cingulum
13. Right fusiform gyrus	30. Left hippocampus
14. Left caudate nucleus	31. Right superior temporal gyrus
15. Left putamen	32. Right precentral gyrus
16. Right Rolandic operculum	33. Left precentral gyrus
17. Right superior orbitofrontal cortex	

unmixing coming from distant signals. The only mixing that remains at the sources is due to the low resolution of the inverse solvers, which is eliminated by the mentioned algorithm used in this work.

2.4.2. Estimation of the primary current at the sources

The EEG scalp signals are projected into the source space in the gray matter using sLoreta (Pascual-Marqui, 2002) as the inverse method because it is easy to calculate and achieves the property of zero localization error in one dipole simulation. In fact, the selection of the inverse method is not a critical step here, since as pointed out above, any linear inverse method would produce the same results after the unmixing procedure (Biscay et al., 2018).

A neonate template was used to approximate the infant's head (Bosch-Bayard et al., 2012), which included gray matter segmentation obtained manually, equivalent to the automatic anatomical labeling (AAL) parcellation of the MNI template (Tzourio-Mazoyer et al., 2002). In Bosch-Bayard et al. (2012), 3 age-dependent templates were created: 2–5, 5–8, and 8–12 months of age. Those templates differed in the size of the head and the skull conductivity. However, for the purpose of this work, we used the template developed for the 5–8 months of age for all subjects, because we needed to have the primary currents estimated at the same sources for all subjects for the purpose of the statistical analysis (see Table 1).

A volumetric grid of 6768 sources was defined, covering all the gray matter, including the thalamus, caudate, insula, and other subcortical structures. These structures were included in the analysis since it has been shown that myelin maturation during the first year of life starts at the subcortical nuclei and gradually spreads to temporal, parietal, occipital, and finally to frontal areas (Baars and Gage, 2013).

Source analysis with the 10/20 system is well-established procedure. Numerous papers point that the accuracy of source imaging increases with the number of EEG electrodes, high-density EEG settings being preferable (Michel and Brunet, 2019; Michel and He, 2019; Nguyen-Danse et al., 2021; Song et al., 2015; Thuraijah et al., 2021). However, a high sensor density is not always possible or desirable and different studies support the use of the 10/20 system for the estimation of EEG sources. For 19 electrodes, the localization error has been estimated from 11 to 40 mm by different authors (Nguyen-Danse et al., 2021; Trujillo-Barreto et al., 2004; Valdes-Sosa et al., 2021; Todaro et al., 2019), depending on the inverse method, the source eccentricity, the forward model, and others. For eLoreta (same as sLoreta) (Todaro et al., 2019) found that with 19 channels, the Point Spread Function (SPF) was in average, of 20 mm in a macaque model. (Michel and He, 2019) concluded that “imperfect spatial sampling precludes source localization and imaging. Some data did suggest that even with a low number of electrodes, we can gain important insight about the underlying brain electric sources by

applying source localization algorithms..., particularly when the precision request remains on the level of larger regions of interest" (as it is the case in this study).

On the other hand, including the Thalamus in this analysis, is supported by several recent papers show that EEG inverse methods adequately recover the activity generated by subcortical structures (Andersen et al., 2020; Attal and Schwartz, 2013; Nahum et al., 2011; Piastra et al., 2021; Pizzo et al., 2019; Seeber et al., 2019).

Estimates for the primary current were obtained for each source and each time point for 24 EEG epochs of 2.56 s for each subject. This information is obtained in the time domain.

2.4.3. EEG effective connectivity source analysis

The effective connectivity between the unmixed signals at the ROIs was calculated by means of the "Isolated Effective Coherence" (iCoh) (Pascual-Marqui et al., 2014). iCoh is a direct and directed measurement of causal information flow in the frequency domain based on Akaike (1968) noise contribution ratio (NCR). In this work, we calculated the iCoh values in the frequency range from 1 to 30 Hz.

The iCoh methodology estimates the partial coherence (values range from 0 to 1) among the signals of a dynamic system using a multivariate autoregressive model. To estimate the information flow from one node to another, all other associations between the nodes are set to zero, except the directional association of interest. In this way, iCoh is asymmetrical, i.e., for each pair of ROIs X and Y, the connectivity from X to Y is generally different than that from Y to X.

For each subject, 24 artifact-free EEG epochs of 2.56 s were processed following the described methodology. The iCoh matrices obtained for the 16 ROIs in the whole frequency range were averaged among segments using the median to protect against possible negative effects from outliers. At the end of the process, one set of iCoh matrices in the frequency domain was stored for each of the 125 subjects.

All calculations in this work were performed by software developed in MATLAB and are publicly available at https://figshare.com/articles/unmixing_test_m/6223778.

2.4.4. Selection of the ROIs for the connectivity analysis

The unmixing procedure already explained limits the maximum number of possible ROIs to be unmixed to 18 since our recordings contain 19 leads. However, to avoid working in the boundaries of matrix singularities and numerical problems, we set this number to 16 ROIs only.

To select the most interesting ROIs for our analysis, we used a data-driven procedure to extract the information from the data instead of an ad hoc selection of those ROIs. For this purpose, we used the eigenvector centrality mapping (ECM) (Lohmann et al., 2010) method to calculate an index of connectivity for each source.

The eigenvector centrality is one of the measurements to assess the node centrality in a network. It favors nodes that high correlations with many other nodes, but especially those nodes connected to other nodes that central nodes to the network themselves, i.e., a node with a high centrality index is not necessarily connected to many nodes but maybe connected to other important nodes. Therefore, it considers the entire connectivity pattern of the network. It has been shown to be a more adequate index to measure the entire connectivity of a network than other measurements like degree centrality or betweenness centrality (Bonacich, 2007; Jayne Bienenstock and Bonacich, 2021; Lohmann et al., 2010).

However, instead of calculating the connectivity index for each subject independently, we used the power method (Del Corso, 1997) to calculate a populational index of connectivity by simultaneously operating over a high number of time signals at the sources. To do that, we concatenated the time signals of all EEG segments of all subjects and processed them simultaneously by using the power method. As a result, a common index of connectivity was calculated for all subjects. This index of connectivity is a global measurement of connectivity obtained

from the whole population simultaneously for each source. We named it as the global connectivity index (GCI).

The power method is an iterative eigenvalue algorithm, which allows obtaining the greatest eigenvalue (in absolute value) of a matrix, in this case, the correlation matrix among the time signal of all sources. The advantage of this simple algorithm is that it is not necessary to calculate the correlation matrix, which can be a very time-consuming process, both in time and memory, or even impossible to calculate for the number of sources and instants of times we deal with. Using this algorithm allowed us processing all the information in only one step and, therefore, obtaining an index which represents the population. The definition of the power method algorithm is provided in the Appendix I.

The GCI was used to determine the 16 regions of interest (ROIs) to be considered for the connectivity analysis based on the sources with the higher connectivity in the brain.

However, using this procedure, the ROIs may not be equally spatially distributed among the whole gray matter. To give a comprehensive overview of the connectivity in the brain, instead of simply selecting the 16 ROIs with the higher GCI, we divided the brain into the following regions: left and right thalamus, left and right temporal, parietal, occipital, and frontal regions; and we selected the sources with the highest GCI inside each of them. Then, we defined a small spherical ROI of 1 cm in diameter around each source. In this way, we guarantee that we cover the whole brain, the best we can, according to our restrictions in the number of ROIs.

Finally, the unmixing algorithm described in Biscay et al. (2018) was applied to the primary currents of the 16 ROIs to determine their unmixed estimates, which were used for the connectivity analysis.

2.4.5. Statistical analysis

Two statistical procedures were used in this study:

- Student's *t*-test versus zero was used to assess the significant iCoh values different from zero. In this case, the mean value of the null distribution was set to 0.1 to avoid accepting iCoh values that were too small between two nodes with nonzero connectivity. The false discovery rate (FDR) (Suckling et al., 2010) criterion was used to correct for multiple comparisons.
- A robust linear regression (robust fit function, MATLAB R2015a) (Holland and Welsch, 1977; Street et al., 1988) was used with the subject's age as the independent factor to determine which iCoh connections changed with age. The age is used as a continuous variable. The robust regressions are less sensitive to outliers and better describe the true tendency of the data. As the descriptive parameter of the age dependency of the iCoh connection with age, we took the regression coefficient accompanying the linear term together with its probability value. A threshold of $p < 0.05$ was used. Again, to correct for multiple comparisons, the FDR criterion was used.

Finally, to facilitate the report of the results, we summarized the information by frequency bands as follows: delta (δ): (1–4 Hz), theta (θ): 4–8 Hz, alpha (α): 8–12 Hz, beta 1 (β_1): 12–20 Hz, and beta2 (β_2): 20–30 Hz.

2.4.6. Data/code availability statement

Data will be sent under reasonable request to the corresponding author and must make appropriate reference to the origin of the data whenever used.

3. Results and discussion

3.1. Global connectivity index

The global connectivity index (GCI) was calculated for all subjects for each of the 6768 sources located in the gray matter and then averaged among subjects. The sources with the highest GCI values were in the subcortical nuclei, as indicated in Error! Reference source not

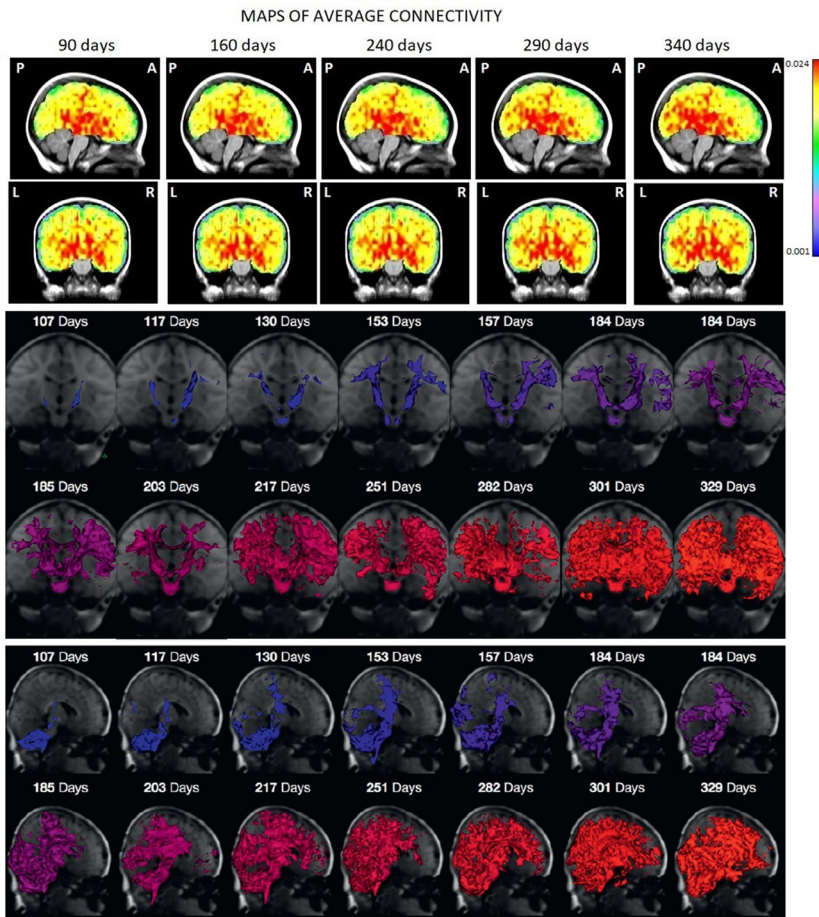


Fig. 1. Top panel, sagittal and coronal maximum intensity (MIP) plots of the GCI values for all sources at different ages during the first year of life. The GCI values for each source and map were calculated by evaluating the specific age in the corresponding regression equation. Bottom panel, Fig. 14.11 in (Baars and Gage, 2013), reproduced here with express permission of Elsevier (License 5,190,240,867,961).

found. These results are highly consistent with the infant's brain myelination process during the first year of life (Baars and Gage, 2013, see Fig. 14.11), included here with permission from Elsevier. According to these authors, the myelination process begins in the pons and cerebellar peduncles and extends to first to the right hemisphere to the optic radiation and the corpus callosum at 3 months old and later continues to the cortex. Finally, myelination occurs in the frontal areas. Our connectivity results also show that the frontal regions exhibited the lowest global connectivity indices.

To visually illustrate the age-related changes in the GCI during the first year of life, we calculated robust linear regressions for each source (robust fit function, MATLAB R2015b), with age as the independent variable. Robust regressions avoid possible negative effects of outliers in the sample, providing a more realistic estimation of the average tendency. In return, we obtained the regression coefficients for each source. Those coefficients were used to evaluate the regression equations for obtaining the mean values of the GCI at 90, 160, 240, 290, and 340 days of age; these values were used to produce images of global brain connectivity at each of the ages. Then, we mapped them into the brain template as sagittal and coronal maximum intensity plot (MIP) maps, which are shown the top panel in Fig. 1. At 90 days, the sources with the maximum connectivity index were concentrated in the subcortical nuclei, with a narrow band appearing around the central sulcus, with a slight preference for the right hemisphere. At 160 days, the GCI started to expand toward the inferior parts of the temporal and parietal lobes and continued expanding to the occipital lobes at 290 and 340 days, covering the whole temporal lobe and the superior parietal gyri. These maps are highly consistent with the map of the infant's brain myelination process during the first year of life (Fig. 14.11 in Baars and Gage, 2013) shown at the bottom panel, printed with Elsevier permission.

Table 2

Brain connectivity for selected regions and the center coordinates in the MNI reference system.

Region Name	ROI Center in MNI Coordinates		
	X	Y	Z
Left Middle Frontal	-38	-2	38
Right Middle Frontal	10	6	38
Left Superior Frontal	-22	-10	48
Right Superior Frontal	10	6	33
Left Precentral	-30	-18	43
Right Precentral	14	-18	38
Left Postcentral	-34	-30	38
Right Postcentral	34	-14	23
Left Inferior Parietal	-50	-30	33
Right Inferior Parietal	30	-34	33
Left Inferior Temporal	-46	-46	-17
Right Inferior Temporal	26	-50	-7
Left Middle Occipital	-42	-66	18
Right Middle Occipital	18	-70	13
Left Thalamus	-14	-14	13
Right Thalamus	-6	-10	3

3.2. ROIs selection

To describe EEG brain connectivity maturation during the first year of life, thus providing a general overview of the whole brain, we divided the sources into 16 areas of the brain and selected one ROI inside each of them around the source with the maximum GCI.

Table 2 shows a list of the selected ROIs and the coordinates of their centroid in the MNI (Montreal Neurological Institute) space. We included the thalamus in this analysis because, first, it receives motor

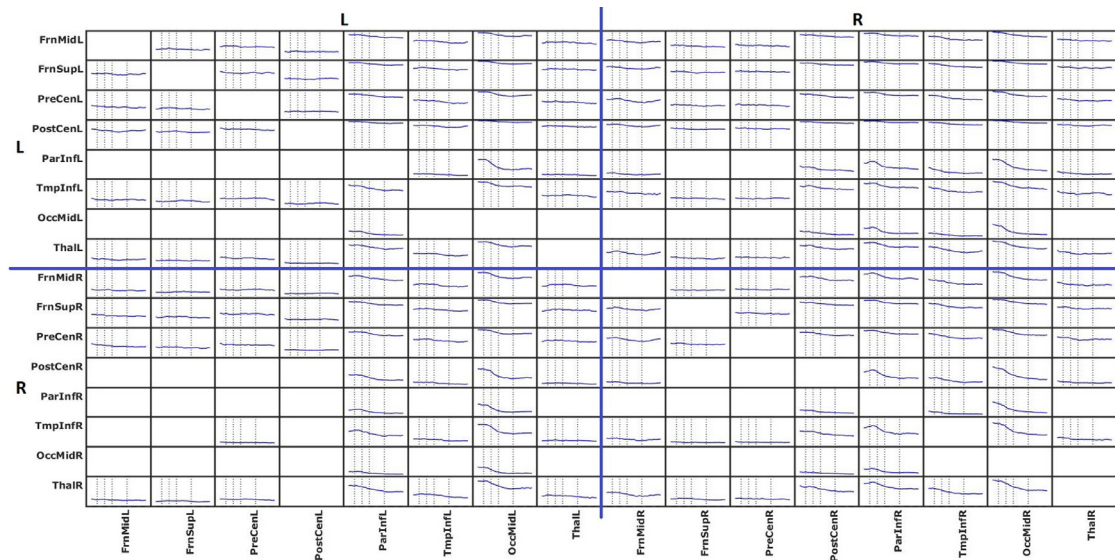


Fig. 2. Isolated effective coherence (iCoh) maps for the 16 ROIs for narrow bands from 1 to 30 Hz. The X-axis is frequency in Hz. The Y-axis represents the iCoh values from 0 to 1. Each box represents the connectivity between the two regions specified in the corresponding row and column. The connections are directed, i.e., the graph is nonsymmetric. The columns in the graph represent the senders, and the rows represent the receivers. The vertical dashed lines indicate the frequencies at 4, 8, 12, 20 and 25 Hz. In each box, the average iCoh values along the frequency axis for the whole sample are plotted. The regions are ordered by the brain hemispheres, first the left and then the right one.

information from the basal ganglia and the cerebellum and receives sensory information from all senses excluding olfaction; second, it projects nerve fibers to the cerebral cortex in all directions and receives back information sent from almost all other regions, behaving like a hub for information exchange. It participates in almost all sensory-motor, consciousness, and alertness processes.

The next step was to calculate the effective EEG brain connectivity (iCoh) among the 16 ROIs.

3.3. Effective connectivity results

The isolated effective coherence (iCoh) (Pascual-Marqui et al., 2014) was calculated among the 16 ROIs for all subjects and averaged in three age groups.

Fig. 2 shows the iCoh values for the 16 ROIs in a tabular shape of 16×16 ROIs for the whole frequency range from 1 to 30 Hz. Each box of the matrix represents the iCoh value from the ROI indicated in the column towards the ROI indicated in the row. The connections are directed, i.e., the graph is nonsymmetric. The ROIs in the matrix columns represent the senders, and the rows represent the receivers. The X-axis corresponds to the frequency, and the vertical dashed lines indicate the frequencies at 4, 8, 12, 20, and 25 Hz. The Y-axis corresponds to the iCoh values from 0 to 1. The blue curve inside each box, are the average iCoh values along the frequency axis, for all infants in the sample. The brain regions have been ordered by hemispheres, first the left and then the right one. This organization allows dividing the figure by the direction of the connectivity in 4 Quadrants:

L→L Quadrant (from the Left to the Left hemisphere)	R→L Quadrant (from the Right to the Left hemisphere)
L→R Quadrant (from the Left to the Right hemisphere)	R→R Quadrant (from the Right to the Right hemisphere)

In the figure, it can be observed that both middle frontal gyri have bilateral connections with the left and right hemispheres. They send connections bilaterally to the superior frontal gyri, precentral gyri, inferior temporal gyri, and thalamus. However, the right middle frontal gyrus also sends information bilaterally to the postcentral and inferior parietal gyri. On the other hand, both middle frontal gyri receive information from all regions. These results corroborate that the right middle

frontal gyrus has been proposed to be a site of convergence of the dorsal and ventral attention networks by serving as a circuit breaker to interrupt ongoing endogenous attentional processes in the dorsal network and reorient attention to an exogenous stimulus (Japee et al., 2015). Newborns have the capacity to orient attention to outstanding stimuli, which is established during the age of 6–12 months (Gomes et al., 2000). Other studies have shown that literacy is associated with the left middle frontal gyrus, whereas numeracy is associated with the right middle frontal gyrus (Koyama et al., 2017).

There were strong connections from the right postcentral gyrus and bilaterally from the inferior parietal, inferior temporal, middle occipital gyri, and thalamus to the frontal lobes. This finding agrees with the anatomical pathways of (Catani and Thiebaut de Schotten, 2008). This observation is also consistent with the fact that the regions of the frontal lobes evaluated mainly govern motor functions. For adequate motor actions, it is essential to have proprioceptive information from the somatosensory area (right postcentral), visual information from the middle occipital, inferior temporal, and a multimodal association area (bilateral inferior parietal) gyrus.

Both superior frontal gyri (SFG) send information bilaterally to the middle frontal, precentral, inferior temporal gyri, and thalamus; the right superior frontal gyrus also sends information to both inferior parietal gyri. Additionally, the superior frontal gyri receive bilateral connections from all areas. The superior frontal gyrus is thought to contribute to higher cognitive functions, mainly working memory (WM), which begins its development during the first year of life (Du Boisgueheneuc et al., 2006). These authors studied patients with left and right frontal lesions and concluded that (i) the lateral and posterior portion of the left SFG is a key component of the neural network of WM, (ii) the highest level of executive processing triggers the participation of this region in WM, and (iii) the left SFG is also involved in spatially oriented processing.

The precentral gyri send connections to the regions of the frontal cortex and the thalamus. However, they both receive connections from all the remaining studied structures. The precentral gyri are well-known primary motor areas responsible for motor development and are the main efferent from the cortex to the medulla; therefore, they must receive information from all cortices.

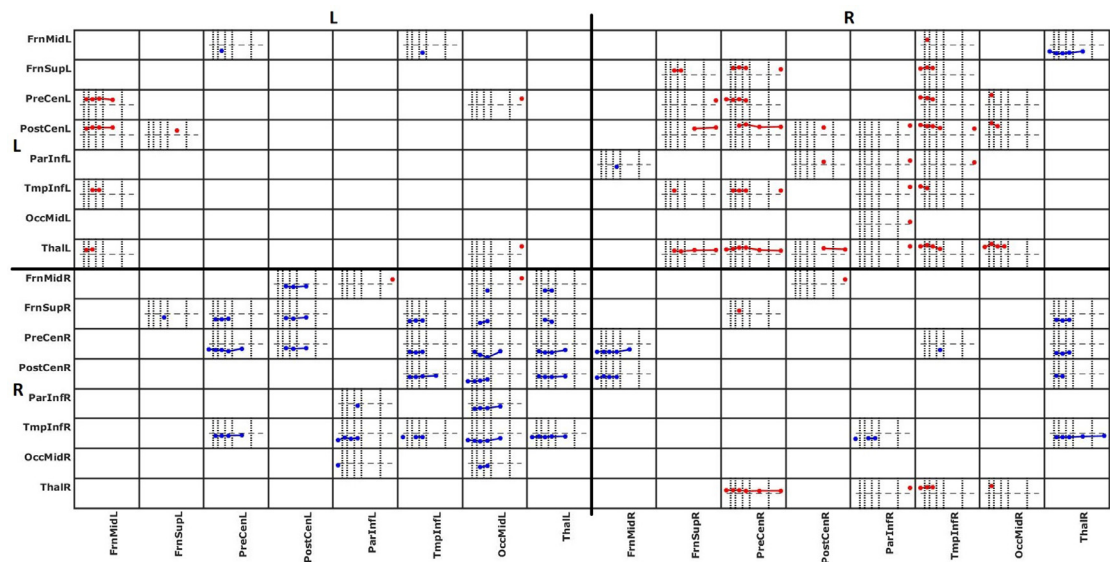


Fig. 3. Age-dependent isolated effective coherence (iCoh) maps for the 16 ROIs summarized by frequency bands. The X-axis is frequency in Hz. The Y-axis shows the coefficients of the age term in a linear regression of the iCohs vs. age for the total sample. A red dot represents a positive coefficient, meaning an increasing connectivity between the two regions with age in the specific frequency band. Blue dots represent negative coefficients, meaning that the connection decreases with age. Each box represents the connectivity between the two regions specified in the corresponding row and column. The connections are directed, i.e., the graph is nonsymmetric. The columns in the graph represent the senders, and the rows represent the receivers. The dashed horizontal line in each box marks the value of the coefficient at 0 and is shown only for reference. The vertical dashed lines indicate the frequencies at 4, 8, 12, 20 and 25 Hz.

Both postcentral regions send information to the frontal lobes. However, only the right postcentral cortex sends connections to the parietal, temporal, and occipital cortices. This result agrees with the predominance of the right hemisphere in infants. The postcentral regions correspond to the somatosensorial cortex that receives communications from the thalamus and the parietal, temporal, and occipital cortices.

The inferior parietal gyri are multimodal association areas. They send information to all other cortices studied and the thalamus. However, they only receive information from the postcentral, temporal, and occipital regions. This area plays a key role in various cognitive functions, including attention, language, visuospatial processing, and action processing, that begin during the first year of life.

The inferior temporal gyri are crucial for visual object recognition and are the last areas to develop in the ventral cortical visual system. The fusiform area, located in the inferior temporal regions, specializes in face recognition, which also starts developing during the first year of age. The inferior temporal regions are very well connected; they send and receive information from the remaining cortices studied and the thalamus.

The occipital regions send information to all studied cortices; however, they only receive information from the bilateral parietal gyri, temporal gyri, and homologous regions of the contralateral hemisphere. Emberson et al. (2017) found surprising evidence that at 6 months, the middle occipital regions, also called the lateral occipital regions, had functional specialization highly similar to adults. These regions track shape information and other object identity cues (e.g., texture/material). This finding provides further evidence of middle occipital gyri specialization from early childhood into infancy, which occurs earlier than developmental trajectories of high-level visual regions.

Fig. 3 shows the iCoh connections that showed a significant dependency on age, summarized by frequency bands (delta, theta, alpha, beta1, and beta2) in the X-axis to facilitate analysis. The Y-axis shows the coefficients of the age term in the robust linear regression of the iCoh values vs. age for the total sample. The red dots represent the positive coefficients, meaning connectivity that increases between the two regions with age in the specific frequency band. The blue dots represent the negative coefficients, meaning that the connections decrease

with age. Each box represents the connectivity between the two regions specified in the corresponding row and column. The connections are directed, i.e., the graph is nonsymmetric, and as shown in Fig. 2, the columns in the graph represent the senders, and the rows represent the receivers. The vertical dashed lines indicate the frequencies at 4, 8, 12, 20, and 25 Hz.

Inspecting Fig. 3 it becomes very apparent that most of the decreasing connections with age are in the L→R quadrant, i.e., from the left to the right hemispheres, as well as most of the increasing connections are in the R→L quadrant, i.e., from the right to the left hemispheres. By columns, it shows that all cortical regions in the right hemisphere (except the right middle frontal area) have increasing connections with age in the theta, alpha, and beta bands. An alternative view of Fig. 3, is shown in Appendix II-B. It facilitates appreciating the age-related maturation aspects by bands.

Connections from the precentral right gyrus to the precentral left gyrus and to the left thalamus, as well as from the right inferior temporal gyrus to the left thalamus, were the only connections in the delta band that increased with age.

The right superior frontal, precentral, inferior parietal, and inferior temporal regions showed increasing high-speed connections in beta2 (from 20 to 30 Hz), mainly to the contralateral hemisphere. In contrast, almost all cortical regions in the left hemisphere (except the middle frontal gyrus) showed decreasing connections with age. Most of the decreasing connections were in theta and alpha, and only one was in the beta2 band (more apparent in the figure in the Appendix II-B).

This brain asymmetry in Fig. 3 is also consistent with other studies. It has been found that the right hemisphere develops first in the ontogenetic development of the brain (Chi et al., 1977). The human brain is right predominant during the preverbal epoch in infants, and this right hemisphere dominance lasts during the first three years of life (Chiron et al., 1997; Schore, 2000). The right hemisphere is understood as an executive regulatory system of the emotional brain involved in inhibitory control (Schore, 2000). In particular, the right orbital prefrontal region specializes in inhibitory control and acts as an executive control for the entire right brain (Garavan et al., 1999; Schore, 2000). This right predominance starts shifting to the left hemisphere by the age

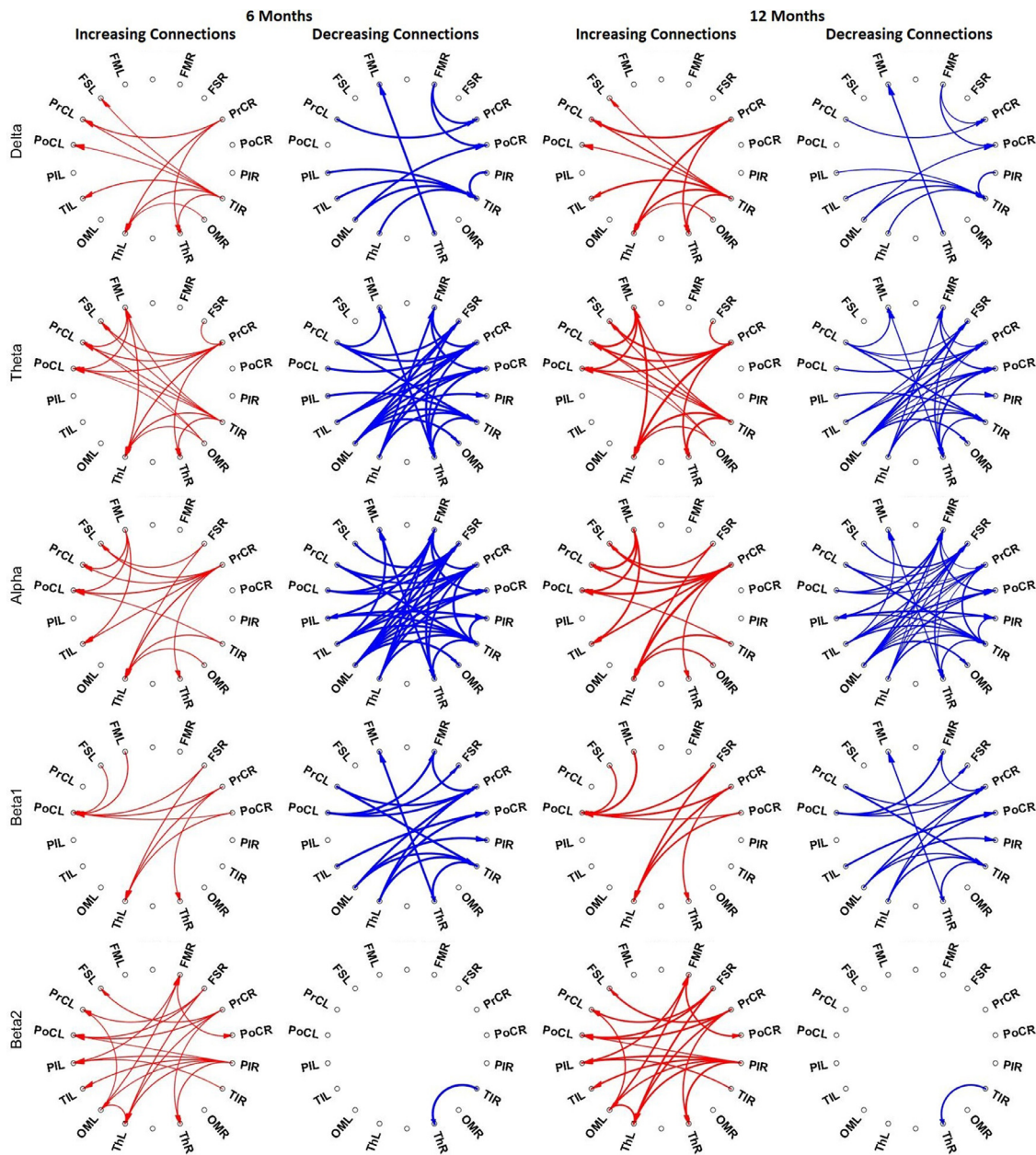


Fig. 4. Maturation of the significant age-dependent isolated effective coherence (iCoh) values during the first year of life. The iCoh regression equations were evaluated at 3, 6, and 12 months. The values at 3 months were assumed to be a baseline and were subtracted from the values at 6 and 12 months to obtain an index of the strength of the connection decrease or increase with age. The results are plotted by frequency bands, by rows. The red arrows show the increasing connections, and the blue arrows show the decreasing connections. The thickness of the lines indicates the strength of the change, and this representation allows us to assess the progression of the change over time. Most of the changes are long-range interhemispheric connections. Many changes in the delta band, especially in the theta and alpha bands, decreased, while all, except one, connections in the beta2 band increased with age. Most of the increasing connections go from the right to the left hemisphere, while almost all decreasing connections go from left to right. This is consistent with the right hemisphere dominance during the first year. Legend: FML/FMR: left/right middle frontal; FSL/FSR: left/right superior frontal; PrCL/PrCR: left/right precentral; PoCL/PoCR: left/right postcentral; PIL/PIR: left/right inferior parietal; TIL/TIR: left/right inferior temporal; OML/OMR: left/right middle occipital; ThL/ThR: left/right thalamus.

of 3 years, when functions initially located in the right hemisphere migrate to the left hemisphere, such as visuospatial and language abilities (Chiron et al., 1997).

The thalamus sends connections to all areas except the middle occipital region. This agrees with the fact that the middle occipital cortex receives information from the primary visual areas. Additionally, all regions sent connections to the thalamus, as expected. Moreover, all connections from the thalamus that change with age exhibit a decreasing trend, consistent with the idea that the thalamus is fully connected at birth, and the nonefficient connections are pruned with age. Interest-

ingly, all decreasing connections are from both Thalamus to the right hemisphere. There is one decreasing connection from the right Thalamus to the left Frontal Middle lobe, which is precisely the only region in the left hemisphere that increases its activity with other structures in the same hemisphere. In contrast, all connections coming to the thalamus are increasing, consistent with the idea that maturation and myelination increase feedback from the cortex to the thalamus.

The definition of frequency bands we have used in the present work may call the attention of some readers because the Alpha peak in the infants appear at slower frequencies, in the range 6–9 Hz (Marshall et al.,

2002). We emphasize that in this work we do not focus on the Alpha peak since our recordings were obtained in the quiet sleep condition, and the Alpha peak is not present. We analyzed the connectivity in the continuous frequency range from 1 to 30 Hz, with frequency bins every 1 Hz, as shown in Fig. 1.

We in fact, performed an additional analysis, redefining the theta and alpha bands as follows: theta (θ): 4–6 Hz, alpha 1 (α_1): 6–9 Hz, and alpha 2 (α_2): 9–12 Hz. In each interval, the upper limit frequency is not reached. The analysis using this bands definition did not show any significant differences: (a) wherever we had significant connections in Theta and Alpha, now we have in theta, alpha 1 and alpha 2; (b) in very seldom cases a significant connection in theta jumped to alpha 1 or vice versa, or appeared in both theta and alpha 1, but with the same tendency; (c) very few connections appeared or disappeared, most probably because of the different bands and the fact that we are augmenting the number of variables in about 20% (from 5 bands to 6 bands), which affected the significant thresholds when corrected by multi comparisons, but the changes are minimal. Not any conclusion or finding changed because of the finer frequency bands analysis. For reference, we present the equivalent figure to Fig. 3 in the Appendix II-A. Therefore, for the sake of simplicity, in the main text we included the analysis of the simpler model using the 5 bands.

Finally, Fig. 4 shows how the connections that change with age during the first year of life. The information is summarized by frequency bands. The iCoh regression equations were evaluated at 3, 6, and 12 months. The values at 3 months were assumed to be a baseline and were subtracted from the values at 6 and 12 months and normalized from 0 to 1 to obtain an index of the strength of the connection decrease or increase with age. The results are plotted by frequency bands, by rows. The positive values (red arrows in the figure) show the increasing connections with age concerning their values at 3 months. In the same way, the negative values (blue arrows in the figure) show decreasing connections with age. The thickness of the lines in the figure indicates the strength of the change; this representation allows us to assess the progression of the change over time.

It can be observed as a general tendency that the decreasing connections are stronger at 6 months, especially in the theta and alpha bands, meaning that the connections tend to decrease at earlier ages. The increasing connections are stronger at 12 months, meaning that the connection tends to continue growing at the end of the first year. Most of the changes are long-range interhemispheric connections.

A table with the exact values of all connections is provided in the Supplementary Materials for more detailed information.

Decreases in connectivity in the delta, theta, and alpha bands were observed at all ages, and they are likely related to the phase of synaptic reduction. This decrease is also observed in the beta1 band. However, the beta2 band mainly showed progressive increases in connectivity with age. Faster connections may be understood as an increase in connection efficiency. Major changes are observed at 6 months of age in relation to the connections at 3 months of age. According to Huttenlocher, (1979) and, Huttenlocher and Dabholkar (1997) that studied the striate cortex from 16 human brains varying in age from 28 weeks gestation to 71 years, two periods were defined, one of rapid synapse production which ends at about postnatal age of 8 months, and a subsequent longer period of synapse elimination which extends past the 3 years of age. This increase in the number of synapses may be related to the increase in connectivity at all frequency bands. However, synaptic reduction was initiated later, which agrees with the decrease in connections from the middle occipital gyri to almost all cortical regions and thalamus in the delta, theta, alpha, and beta1 bands. Concerning the increases, major changes are observed in those connections that originate in the right hemisphere. As we have mentioned before, the maturation of an infant's brain begins in the right hemisphere, and at approximately 3 years old, functions such as language are transferred to the left hemisphere. Major increases in connectivity were found in interhemispheric connections, as well as connectivity from the left and right thalamus with all

other cortical regions. These changes agree with the periods of synaptogenesis of the cortical regions present from 3 months of age and with myelination periods, particularly with the increase in size of the corpus callosum which explains the major increases of interhemispheric connections (Sakai et al., 2017). They are also consistent with the view that increasing cognitive capacity during childhood may coincide with a gradual loss rather than the formation of new synapses and presumably a strengthening of remaining synaptic connections (Thompson and Nelson, 2001).

4. Conclusions

The results obtained support our hypothesis. EEG connectivity between different cortical areas showed age-related changes during the first year of age, increasing in all frequency bands during the periods of myelination and synaptogenesis and decreasing in all frequency bands during the period of synapse reduction. All these changes showed pre-dominance in the right hemisphere.

The connectivity results analyzed in this paper have been obtained from a cross-sectional sample of infants during the first year of life. It would be interesting to corroborate them in a longitudinal sample, however, it should be noted the difficulties of getting the collaboration from the parents to keep a sample of healthy infants during a whole year being administered a full set of neurological evaluations periodically.

Credit authorship contribution statement

Jorge Bosch-Bayard: Formal analysis, Writing – original draft, Writing – review & editing. **Rolando J. Biscay:** Formal analysis, Writing – original draft, Writing – review & editing. **Thalia Fernandez:** Visualization, Writing – original draft, Writing – review & editing. **Gloria A. Otero:** Visualization, Writing – original draft, Writing – review & editing. **Josefina Ricardo-Garcell:** Writing – original draft, Writing – review & editing. **Eduardo Aubert-Vazquez:** Formal analysis, Writing – original draft, Writing – review & editing. **Thalia Harmony:** Methodology, Formal analysis, Writing – original draft, Writing – review & editing.

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Author(s') disclosure statement(s) (actual or potential)

Nothing to declare.

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Data/code availability statement

Data will be sent at the written request of authors who must refer to the origin of the data.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.neuroimage.2022.119035](https://doi.org/10.1016/j.neuroimage.2022.119035).

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